Serag Eldien Mahmoud

Backend / Full-Stack Software Engineer | Node.js · NestJS · TypeScript · PostgreSQL · AWS
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SUMMARY

Backend / Full-Stack Software Engineer with 2+ years of hands-on experience building and deploying production web applications with **Node.js, NestJS, and TypeScript**. Experienced in REST API design, authentication, payment integrations, database schemas, Docker deployments, and AWS. Built and shipped 5+ client applications end to end, including a live e-learning platform with 1,000+ active users, plus experience in Spring Boot and AI-powered backends.

SKILLS

Programming Languages: TypeScript, JavaScript, Java, C++, SQL
Backend: Node.js, NestJS, Spring Boot, REST API design, RabbitMQ, BullMQ, Jest
Databases: PostgreSQL, MongoDB, MySQL, Redis, Prisma ORM, TypeORM
Cloud & DevOps: Docker, AWS (EC2, S3), GitHub Actions (CI/CD), Nginx, Linux
Frontend: React, Tailwind CSS, TanStack Router, Zustand
Developer Tooling: Git / GitHub, Swagger, Cursor, Claude Code (AI-assisted development)
Engineering: Clean Architecture, SOLID, Design Patterns, Payment Integration, OOP, Unit Testing, Agile, DSA (220+ LeetCode)

EXPERIENCE

Freelance Software Engineer — Self-Employed (Remote) Jan 2024 – Present

Designed, built, and shipped 5+ full-stack web applications for clients end to end.

- Delivered production backends for 5+ client projects using **NestJS, React, PostgreSQL, MongoDB, and AWS**, accelerating feature delivery through modular architecture and AI-assisted workflows.
- Eliminated client-reported auth issues by implementing **JWT, OAuth 2.0 (Google/Facebook via Passport), OTP, Clerk, and RBAC** across all projects.
- Designed relational and document schemas for read-heavy workloads; applied indexing and query optimization that improved query performance on the Englishom platform.

AI & ML Intern — IMPACT, Zewail City of Science and Technology Jan 2023 – Sep 2023

- Built an automated document-processing pipeline in **Python with OpenAI APIs** to extract structured data from PDFs; it became the foundation for the DocTalker graduation project.

Problem-Solving Instructor — Microsoft Student Partner (MSP) Sep 2021 – Jun 2022

- Taught Problem-Solving Levels 0 and 1 in **C++**, covering core algorithms and data structures for university students.
- Mentored 40+ students through structured weekly problem sets and mock contests on **vJudge**, tracking progress and preparing them for exams.

PROJECTS

Englishom — Live Online Courses Platform (Client Project) — 1,000+ active users

englishom.com | github.com/itzSerag/Englishom · NestJS, TypeScript, MongoDB, Docker, AWS S3, SES, Paymob, Passport OAuth

- Built the full **NestJS + TypeScript** backend for a live e-learning platform with **1,000+ active users**, using a modular monolith.
- Integrated OTP (Brevo), Google/Facebook OAuth (Passport), and the **Paymob** payment gateway, covering the complete authentication and purchase lifecycle.
- Served course content through **AWS S3 + CDN** for scalable, fast media access.

DocTalker — AI RAG Documents Chat Platform

Live Demo | github.com/itzSerag/DocTalker · Node.js, Express.js, TypeScript, MongoDB, OpenAI API, Gemini, AWS S3, Stripe, Google OAuth, React, Tailwind

- Built a **RAG backend** supporting PDFs, folders of documents, YouTube URLs, web scraping, and handwritten uploads (including Arabic), with chunk embedding via Hugging Face and context-aware Q&A using OpenAI and Google Gemini.
- Implemented a tiered subscription system with **Stripe** (checkout sessions, webhook-driven access control, per-tier limits).
- Built the responsive React (Vite / TypeScript) frontend and its UI design system.

Chatty — Full-Stack Real-Time Chat Application

Live Demo | github.com/itzSerag/Chatty · NestJS, MongoDB, PostgreSQL, Socket.IO, Cloudinary, React (Vite), Tailwind CSS, Docker Compose

- Implemented the repository pattern in NestJS (MongoDB); built real-time chat and live presence with **Socket.IO**.
- Built a multi-image upload flow (up to 5 images at once) to **Cloudinary**, saving references in PostgreSQL.
- Built the React + Tailwind frontend with Zod-validated forms, plus centralized logging and exception handling.
- Containerized backend and PostgreSQL as separate services with **Docker Compose** (Alpine images); deployed on Render and Vercel.

EDUCATION & CERTIFICATIONS

B.Sc. in Computer Science — Ain Shams University, Cairo, Egypt Sep 2020 – Sep 2024

- Coursework: Databases, Algorithms & Data Structures, OOP, Parallel Programming, AI Intro. Competed in **ECPC** 2022.

Certifications: IBM Web Development Professional Training | **IBM Software Developer Training** (Credly) **| Certifications & References**
Languages: Arabic (Native) · English (Professional Working Proficiency)